



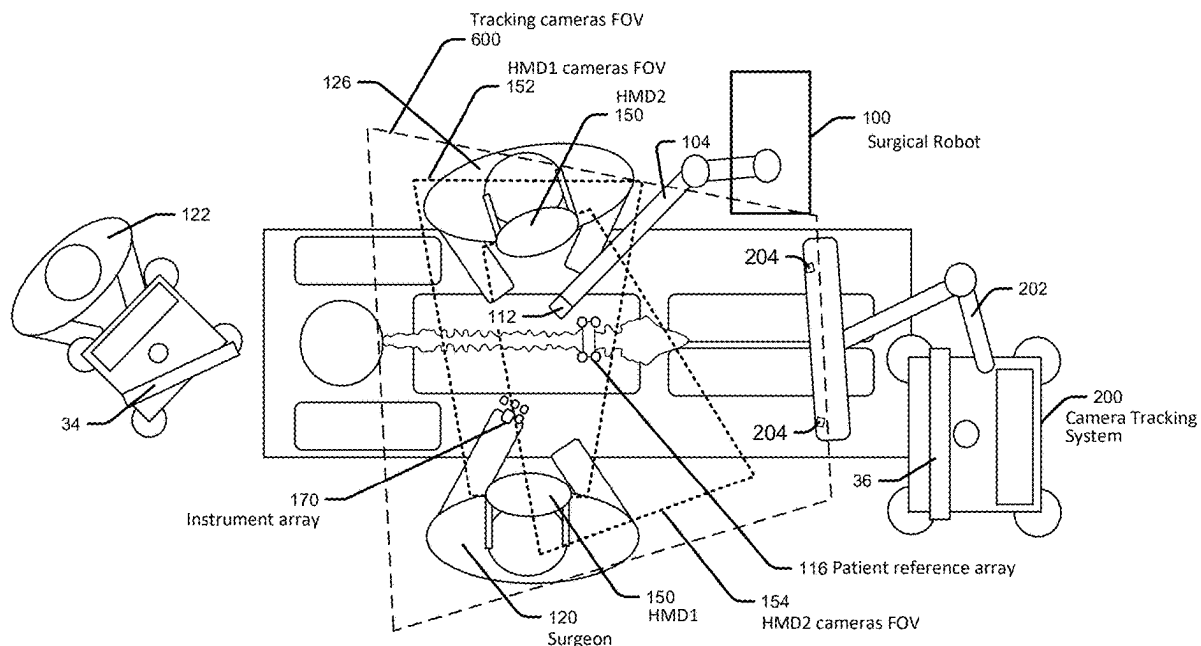
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Calloway et al.(10) **Pub. No.: US 2025/0281244 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **CAMERA TRACKING SYSTEM FOR
COMPUTER ASSISTED SURGERY
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(2013.01)

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ABSTRACT

A camera tracking system for computer assisted navigation during surgery. Operations identify locations of markers of a reference array in images obtained from tracking cameras imaging a real device. Operations determine measured coordinate locations of a feature of a real device in the images based on the identified locations of the markers and based on a relative location relationship between the markers and the feature. Operations process a region of interest in the images identified based on the measured coordinate locations through a neural network configured to output a prediction of coordinate locations of the feature in the images. The neural network has been trained based on training images containing the feature of a computer model rendered at known coordinate locations. Operations track pose of the feature of the real device in 3D space based on the prediction of coordinate locations of the feature of the real device in the images.



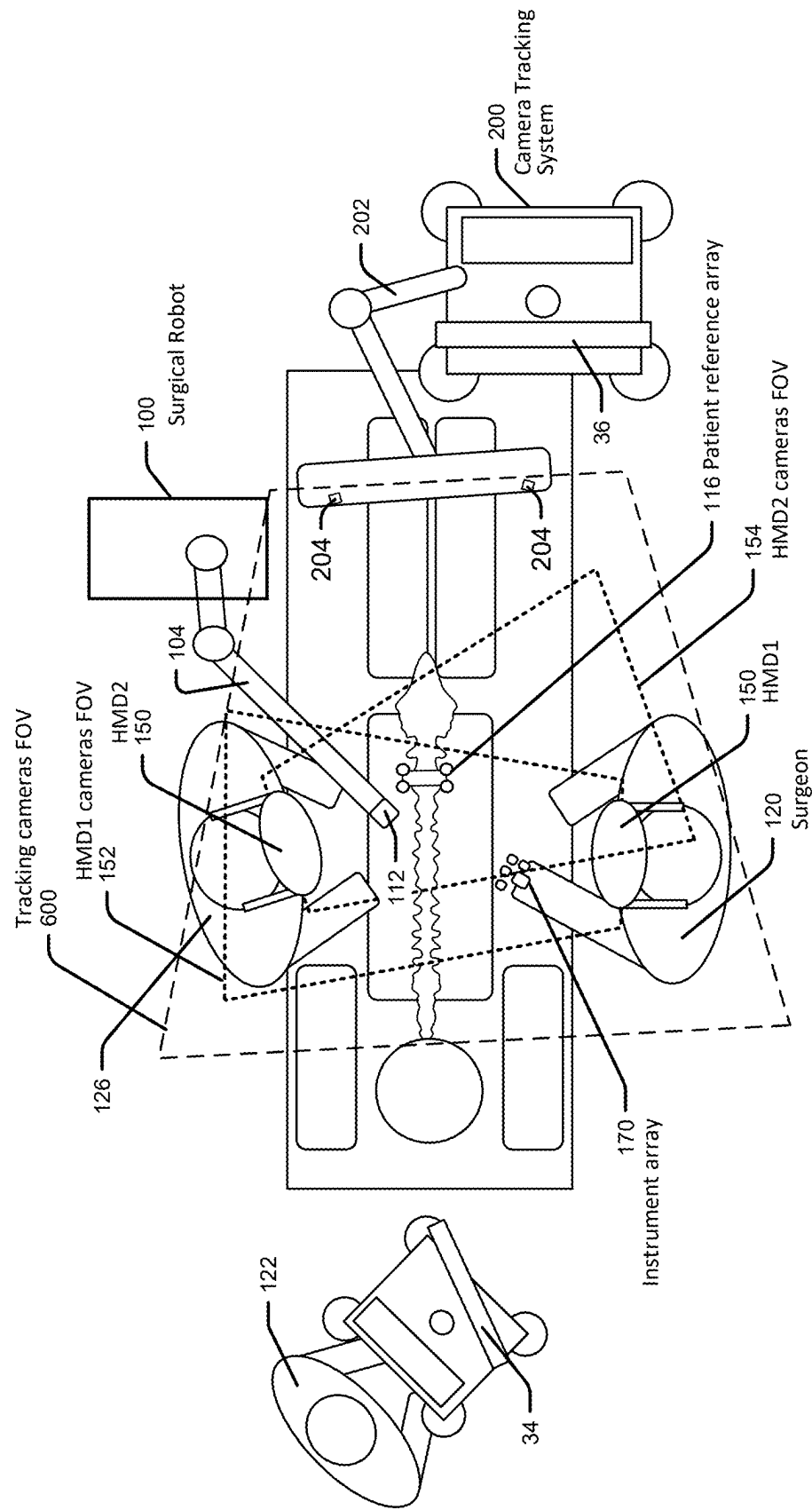


Fig. 1

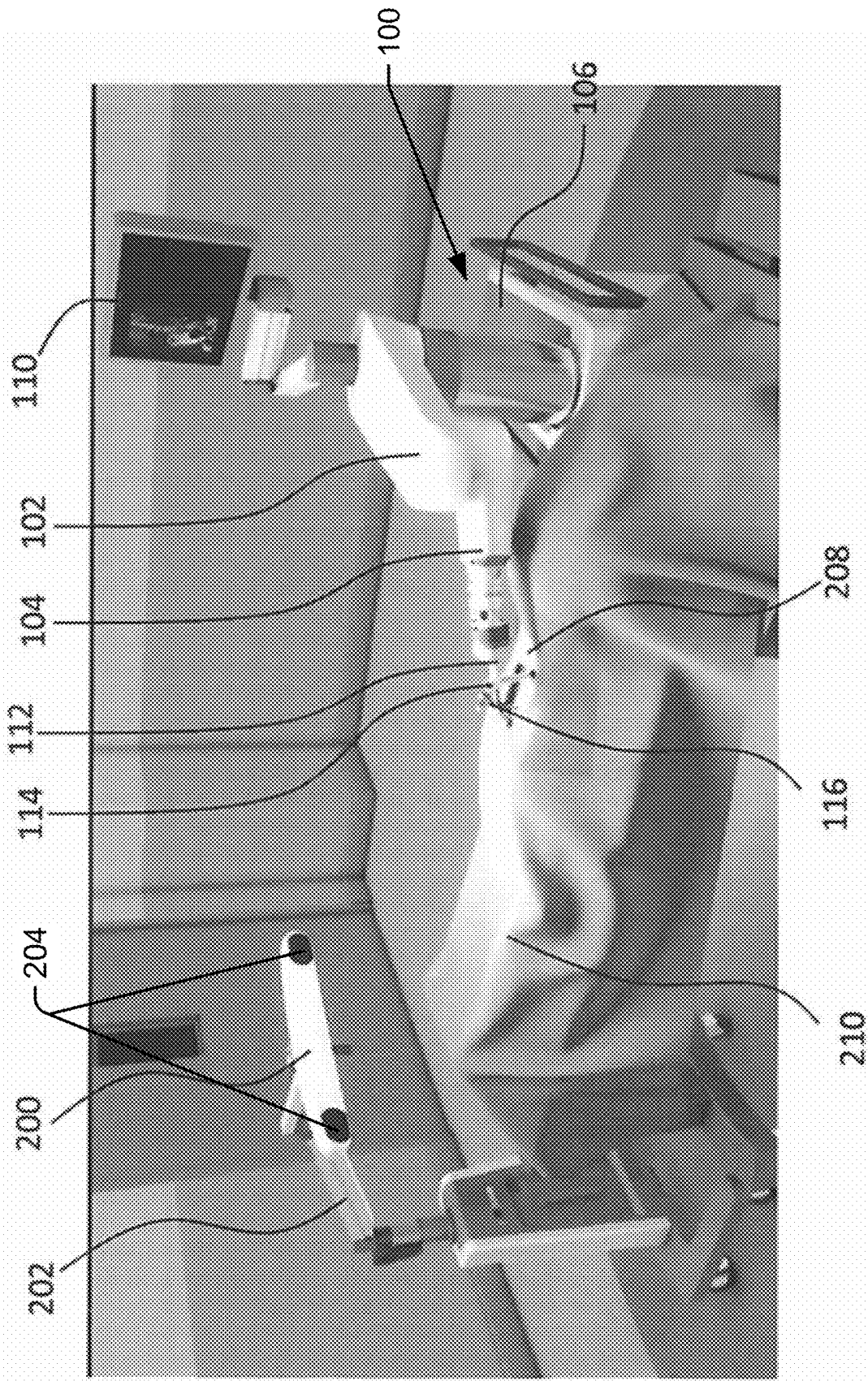


Fig. 2

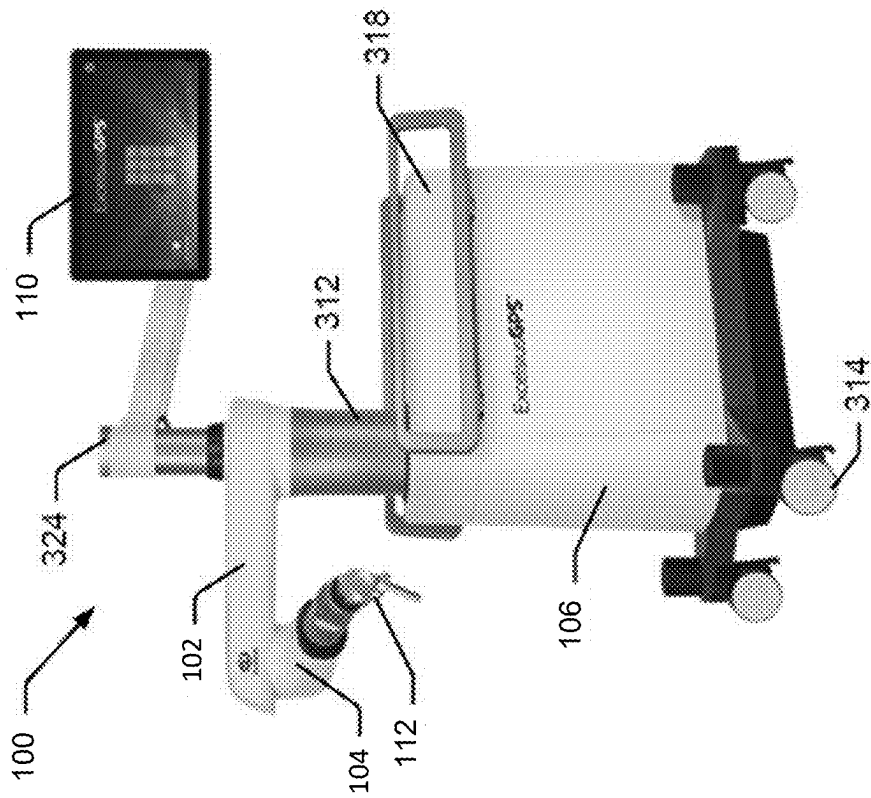
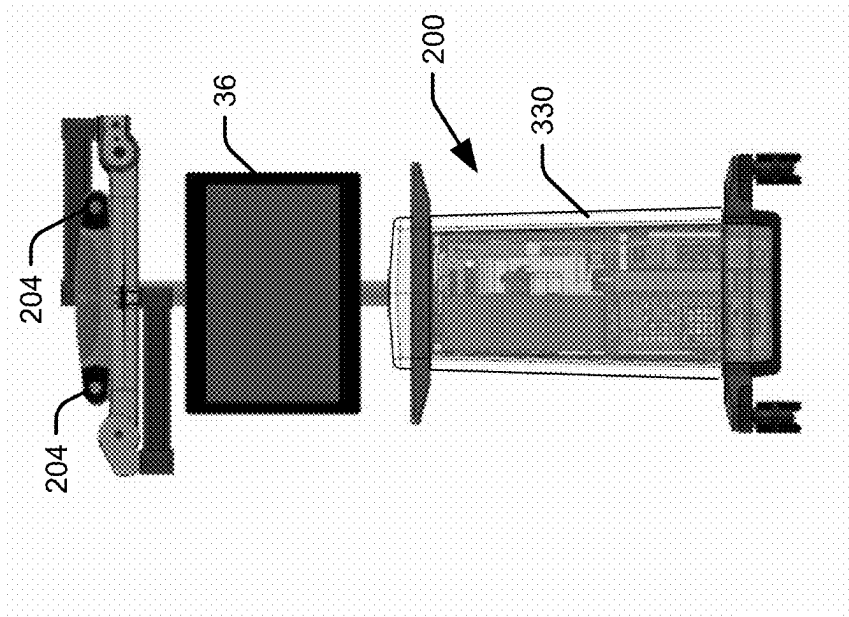


Fig. 3

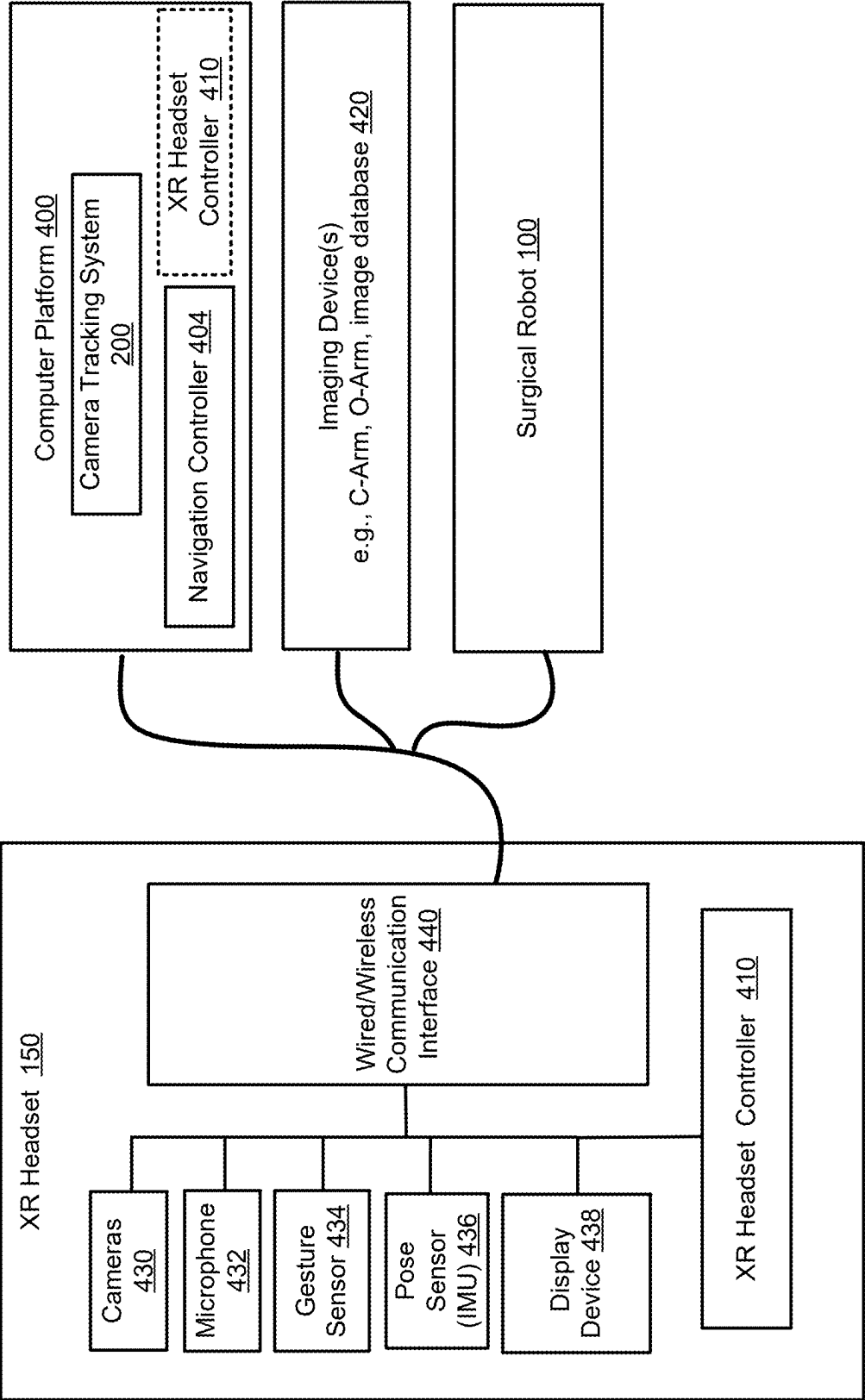


Fig. 4

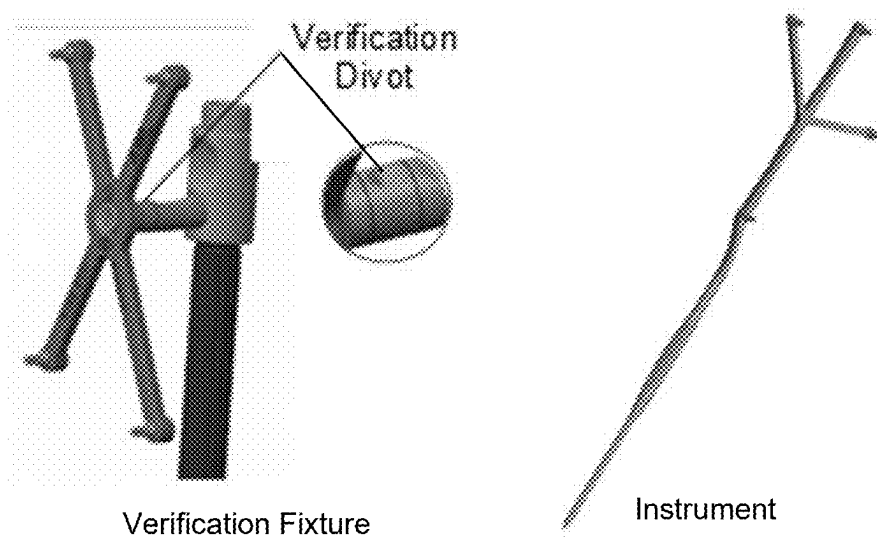


Fig. 5

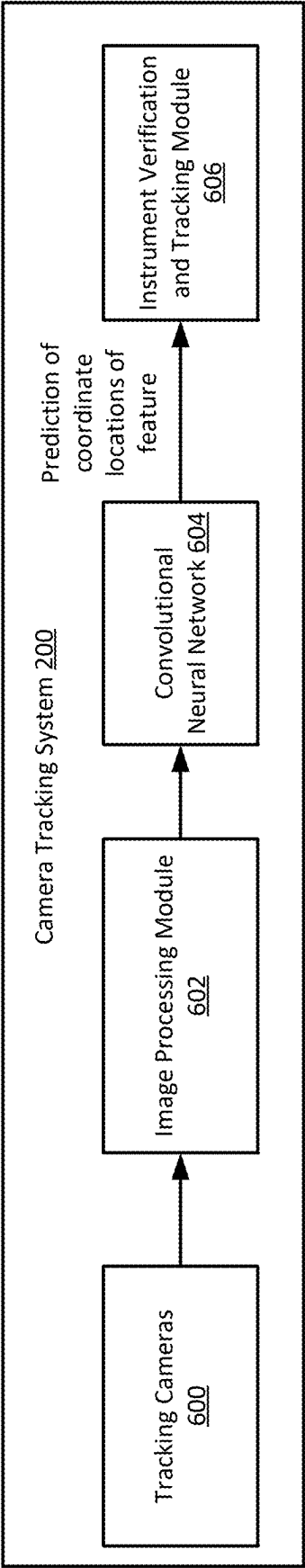
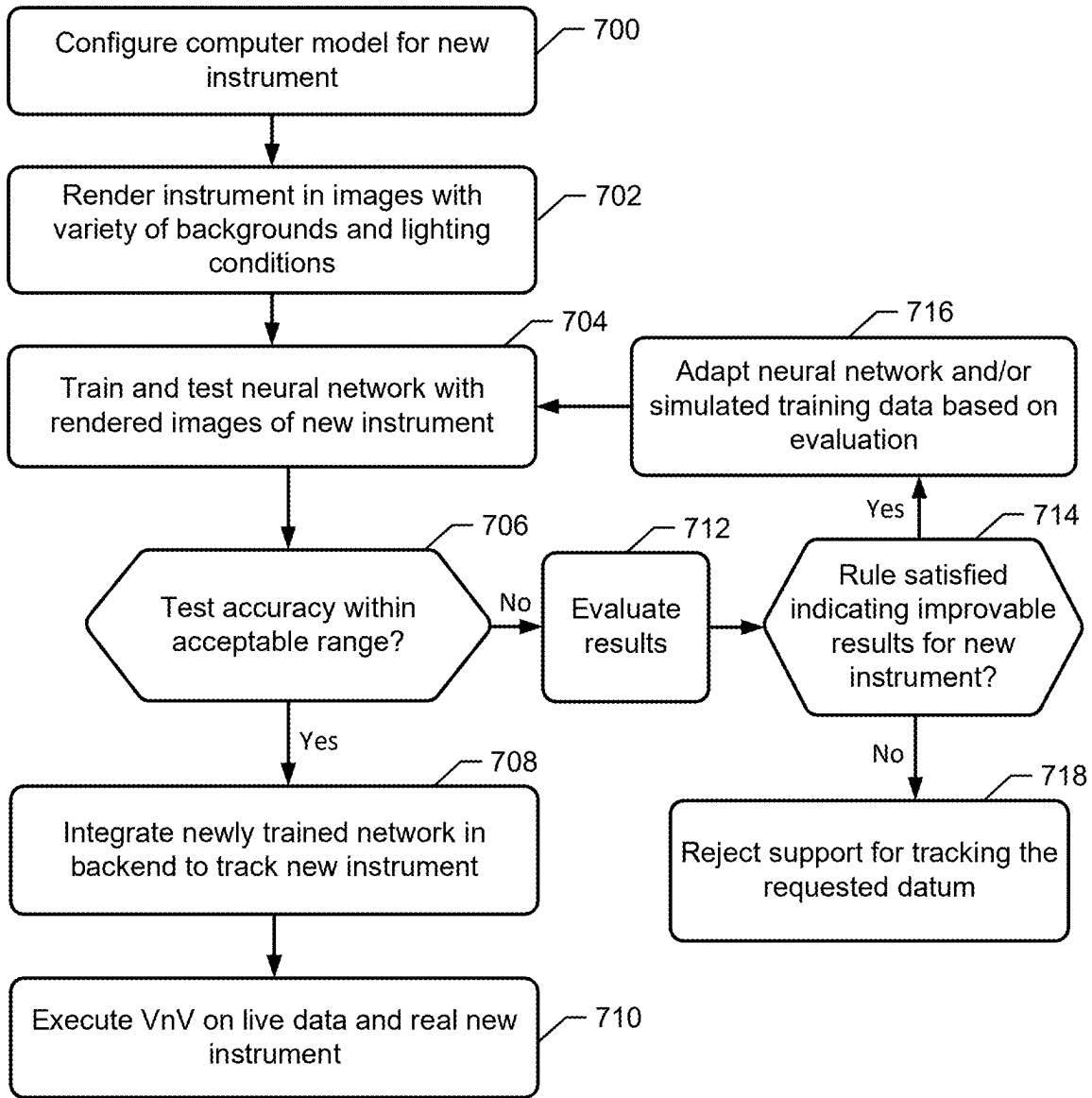


Fig. 6

**Fig. 7**

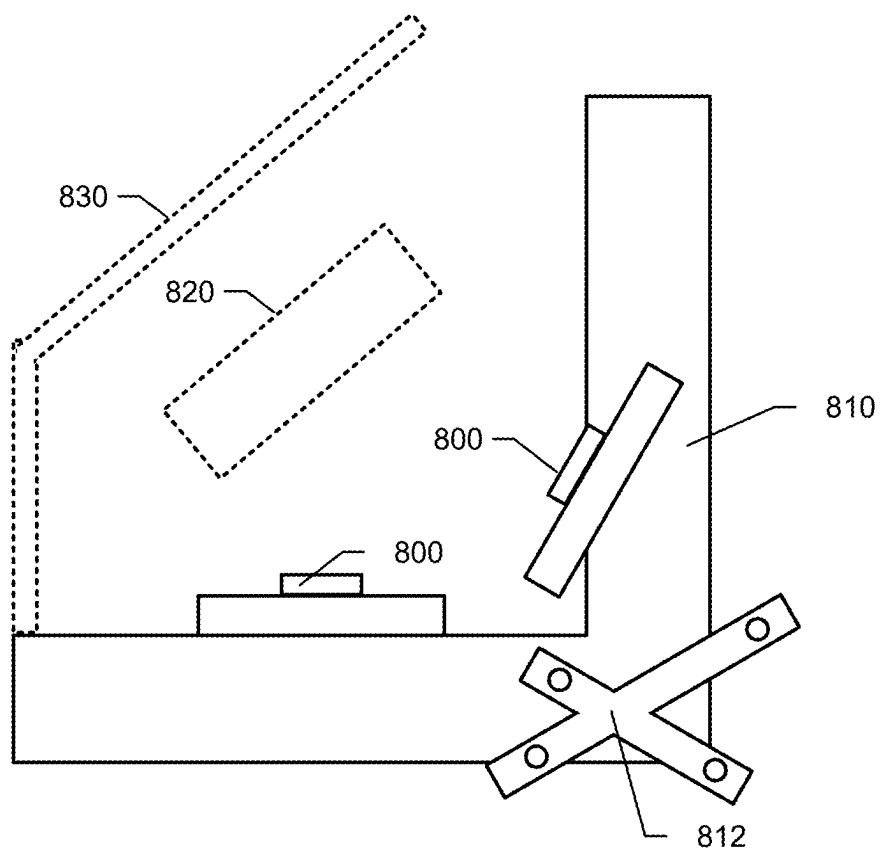


Fig. 8

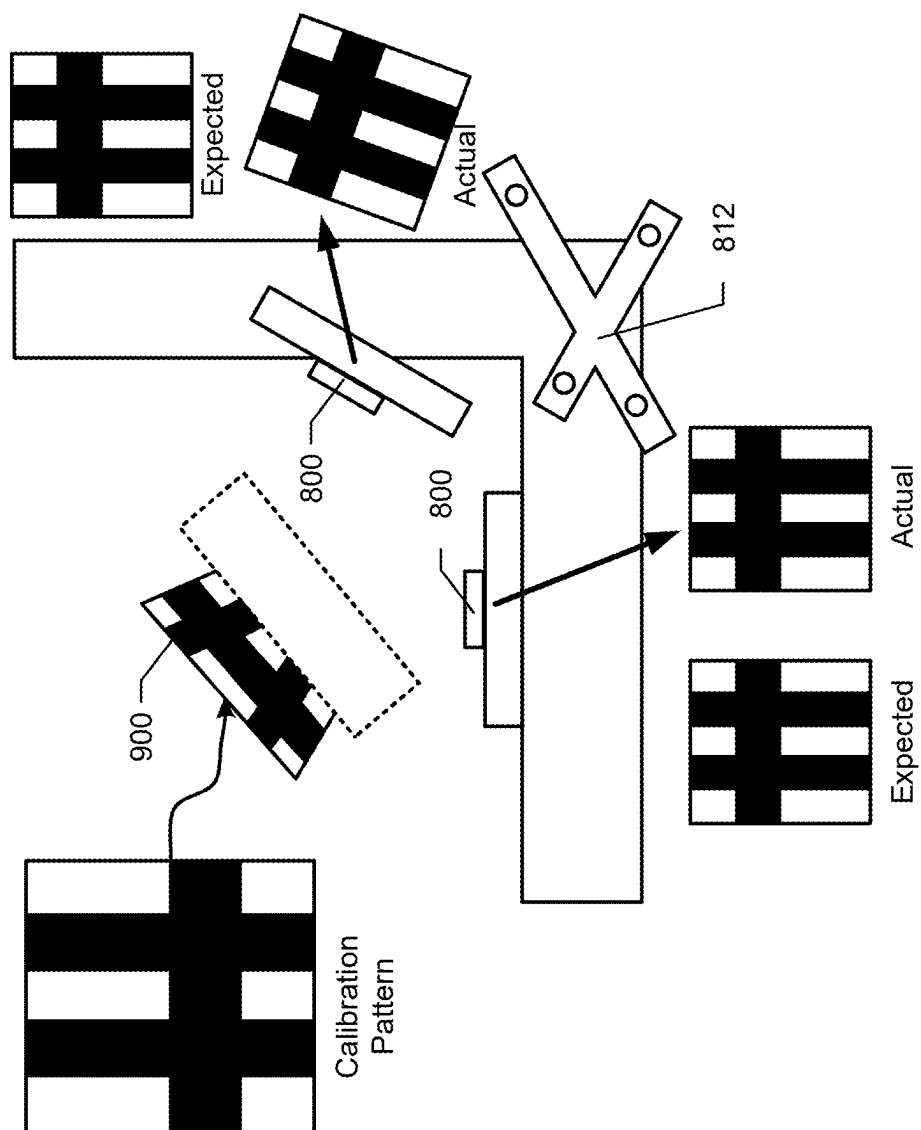


Fig. 9

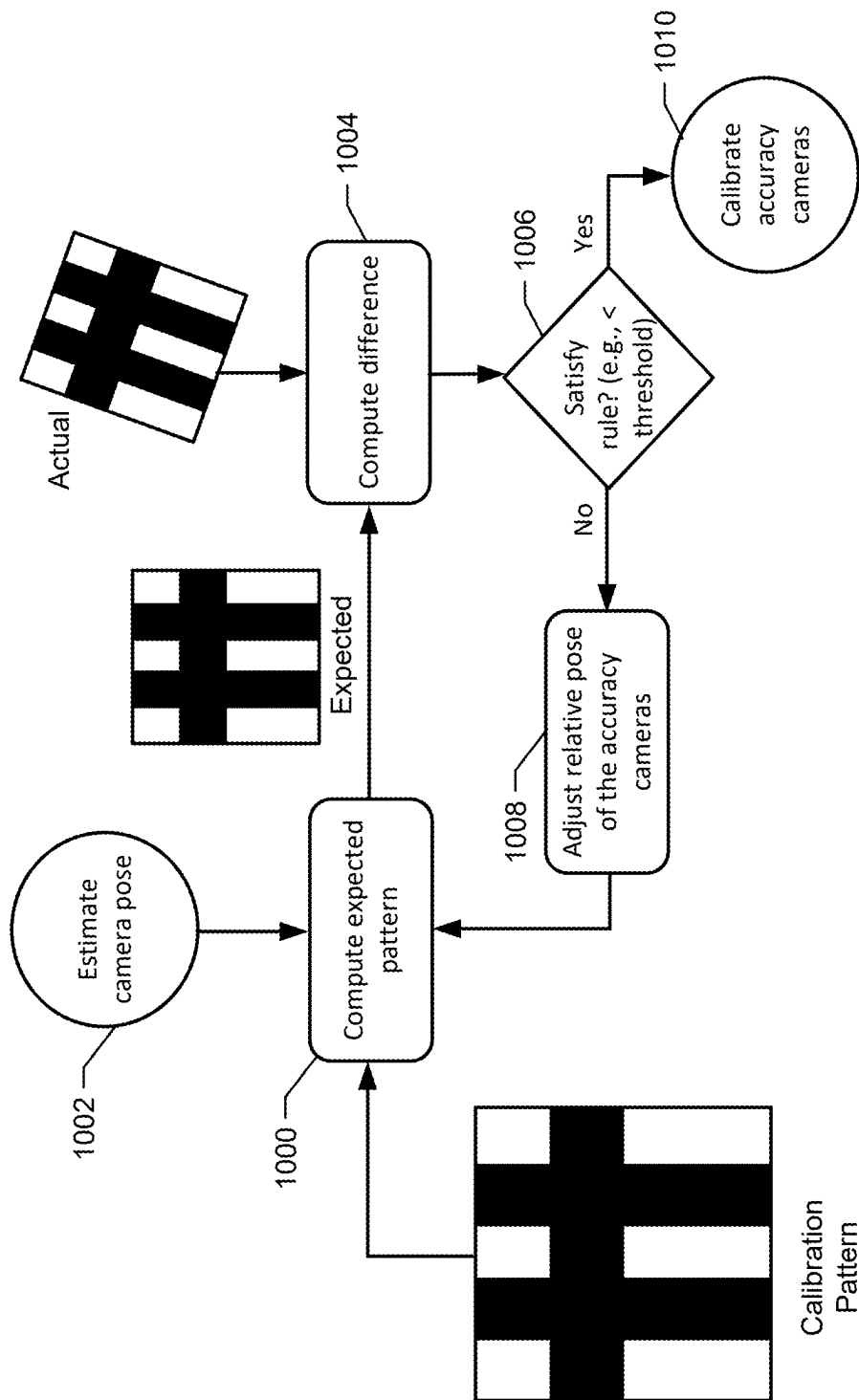


Fig. 10

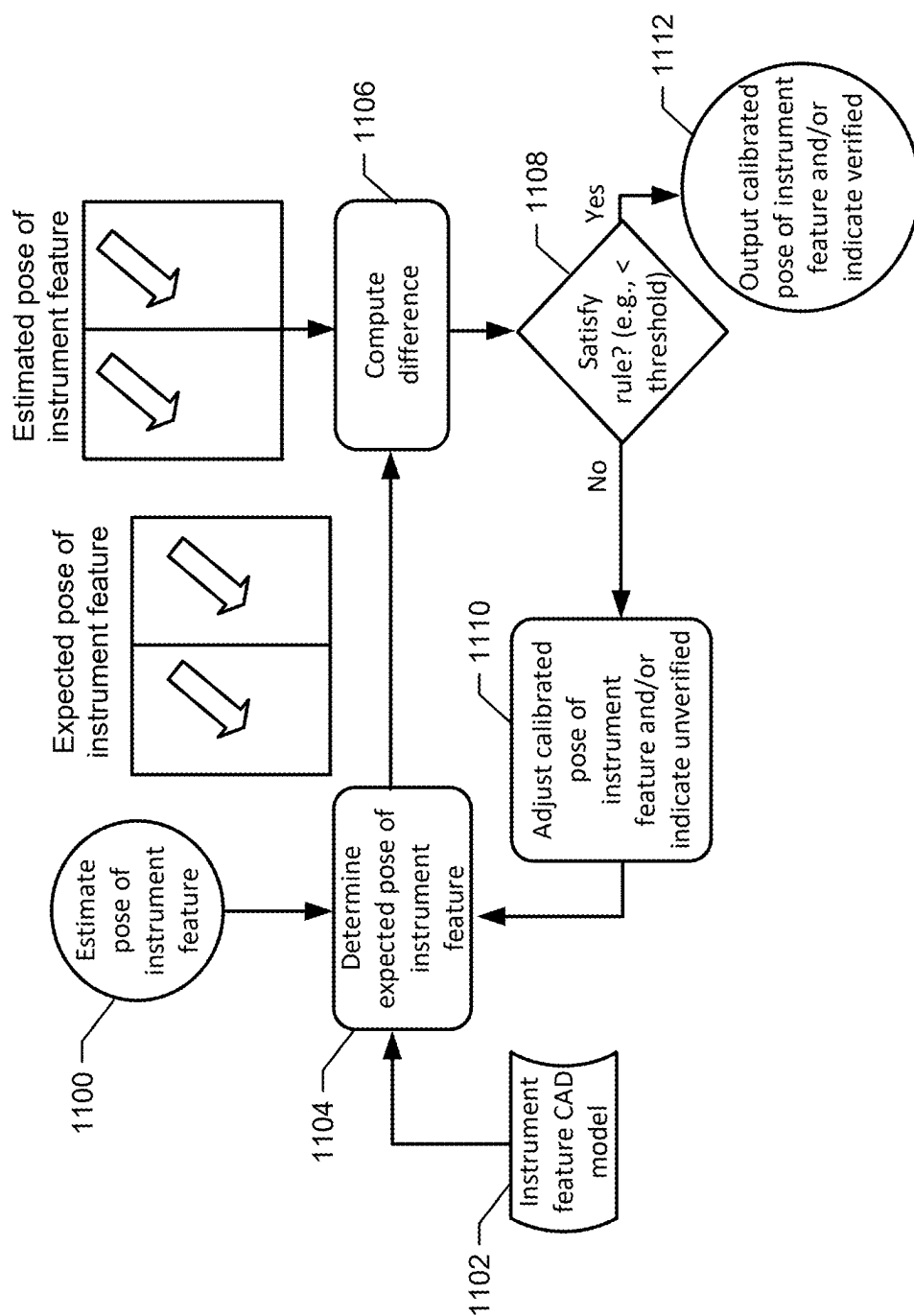


Fig. 11

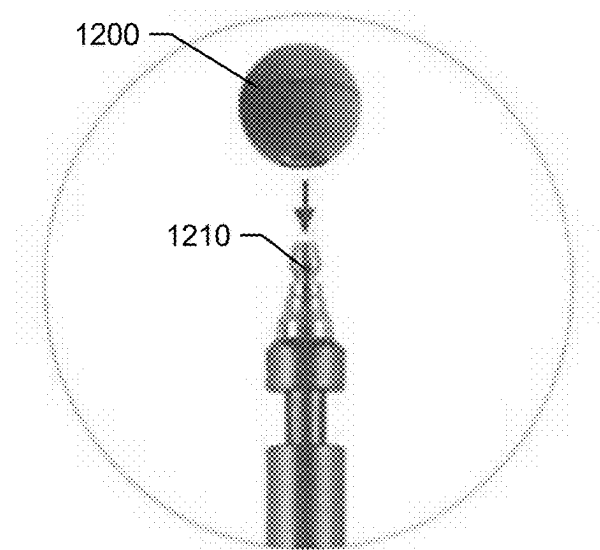


Fig. 12

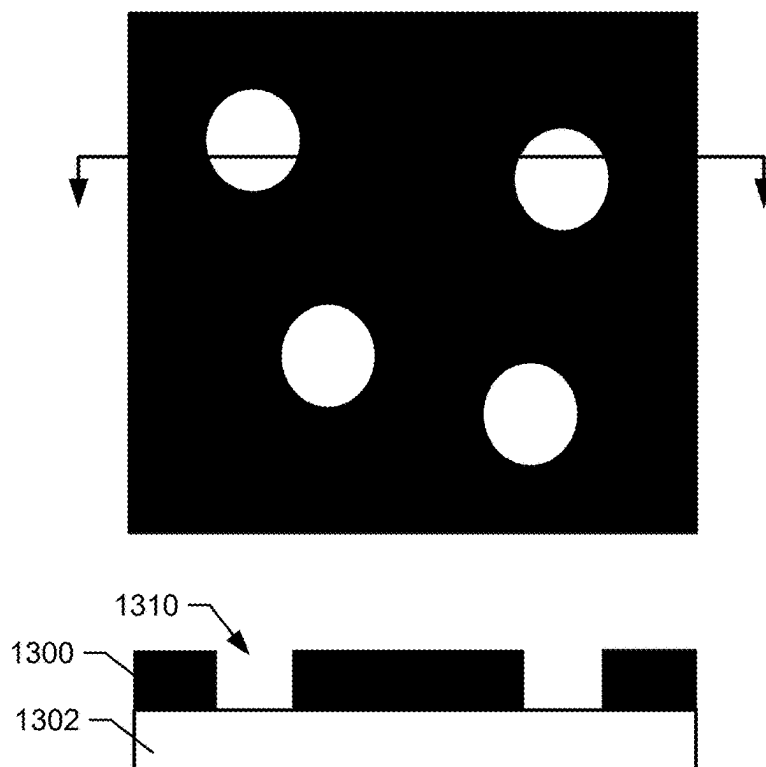


Fig. 13

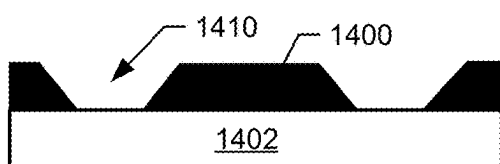


Fig. 14a

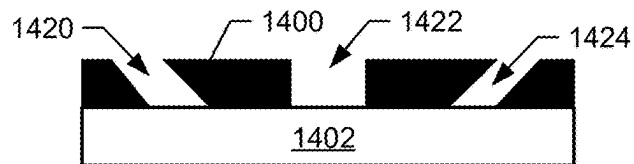


Fig. 14b

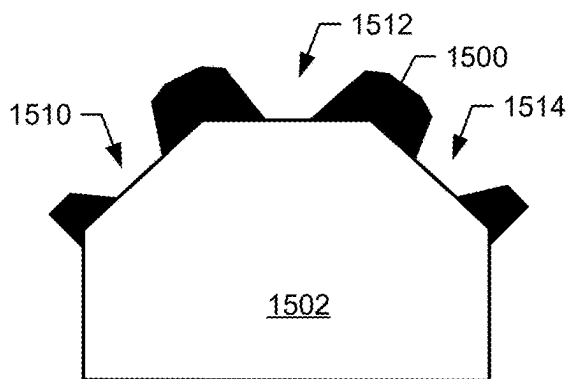


Fig. 15a

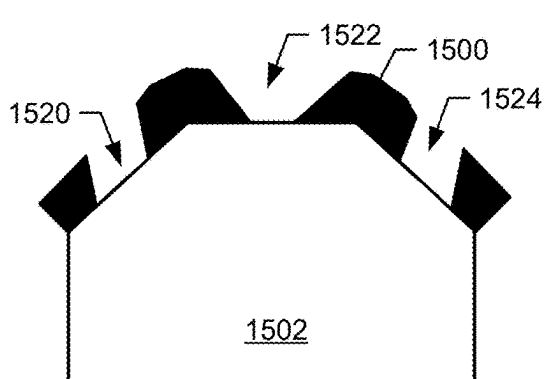


Fig. 15b

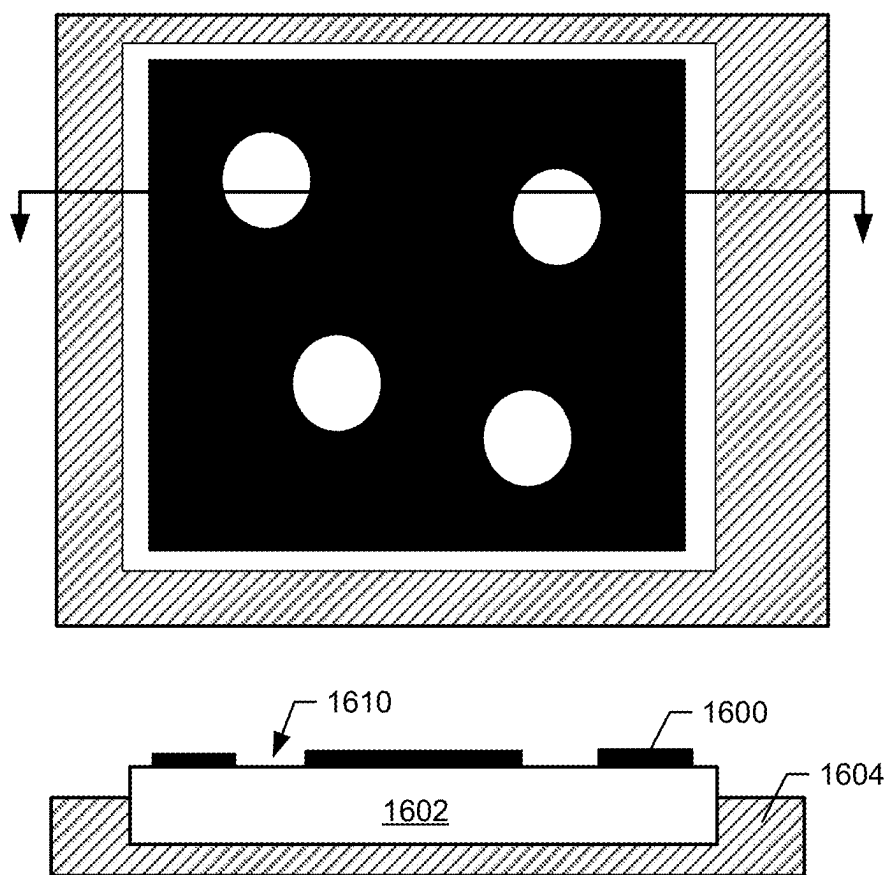


Fig. 16

CAMERA TRACKING SYSTEM FOR COMPUTER ASSISTED SURGERY NAVIGATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional application of U.S. patent application Ser. No. 18/045,474 filed on Oct. 11, 2022, which is incorporated in its entirety herein.

FIELD

[0002] The present disclosure relates to medical devices and systems, and more particularly, camera tracking systems used for computer assisted navigation during surgery.

BACKGROUND

[0003] A computer assisted surgery navigation system has become a well-established technique in an operating room for provide a surgeon with computerized visualization of how a surgical instrument or other device that is posed relative to a patient correlates to a pose relative to medical images of the patient's anatomy, and how those poses correlate to a pre-operative surgical plan. Camera tracking systems for computer assisted surgery navigation typically use a set of tracking cameras to track pose of a reference array on the surgical instrument, which is being positioned by a surgeon during surgery, relative to a patient reference array (also "dynamic reference base" (DRB)) affixed to a patient. A computer model of a real instrument is associated with a reference element, so that the computer model can be overlaid on registered images of patient's anatomy. The camera tracking system uses the relative poses of the reference arrays to determine how the real instrument is posed relative to a patient and to determine how the computer model of the real instrument is to be correspondingly posed as on overlay on the medical images. The surgeon can thereby use real-time visual feedback of the relative poses to navigate the surgical instrument during a surgical procedure on the patient.

[0004] To ensure fidelity of the instrument computer model overlaid on the medical images, accuracy of the real instrument relative to the computer model needs to be verified prior to use. The accuracy check can be performed by bringing the tip of the tracked real instrument into a divot associated with another reference element. The divot is typically a cone-shaped depression. FIG. 5 illustrates an instrument that can be tracked by the camera tracking system as its tip is touched to a verification divot on an arm supporting a tracked reference array which is also tracked by the camera tracking system. The predicted position of the instrument tip determined based on the instrument computer model is compared with the measured position determined based on the tracked position of the divot. Assuming the user has properly positioned the instrument tip in the divot, the distance between the predicted and measured positions determines the accuracy of the tracked instrument. If the accuracy check does not pass, that instrument is not verified and should not be used.

[0005] Example sources of inaccuracies during instrument verification include:

- [0006]** 1. Deformed instrument, e.g., bent tip;
- [0007]** 2. Deformed reference element, e.g., bent reference array fixture or bent marker mounting post, where

it is noted that a relatively small angular shift in the reference element can result in corresponding very small error for tracking of the reference element, but result in a much larger error for tracking the instrument tip; and

[0008] 3. Inaccuracies in optical markers due to manufacturing defects, smudges, or inaccurate mounting of optical markers on mounting posts or within marker mounting sockets.

[0009] Verification of the instrument at time of use can enable compensative for inaccuracies and improve the fidelity of tracking so that the measured location of the physical tip matches the model-based prediction.

SUMMARY

[0010] Some embodiments of the present disclosure are directed to a camera tracking system for computer assisted navigation during surgery. The system includes at least one processor configured to perform operations that include to identify locations of markers of a reference array in a set of the images obtained from tracking cameras imaging a real device with at least partially overlapping field-of-views. The operations determine measured coordinate locations of a feature of a real device in the set of the images based on the identified locations of the markers and based on a relative location relationship between the markers and the feature. The operations process a region of interest in the set of the images identified based on the measured coordinate locations through a neural network configured to output a prediction of coordinate locations of the feature in the set of the images. The neural network has been trained based on training images containing the feature of a computer model rendered at known coordinate locations. The operations track pose of the feature of the real device in three-dimensional (3D) space based on the prediction of coordinate locations of the feature of the real device in the set of the images.

[0011] Some other embodiments of the present disclosure are directed to another camera tracking system for computer assisted navigation during surgery. The system includes at least one processor configured to perform operations that include to obtain images from accuracy cameras imaging a calibration pattern with at least partially overlapping field-of-views, wherein the accuracy cameras are connected to a reference array of markers through a rigid fixture. The operations identify an actual calibration pattern in the images from the accuracy cameras. The operations compute an expected calibration pattern in the images from the accuracy cameras based on a relative pose of the accuracy cameras. The operations adjust the relative pose of the accuracy cameras based on difference between the actual calibration pattern and the expected calibration pattern in the images from the accuracy cameras.

[0012] Some other embodiments of the present disclosure are directed to another camera tracking system for computer assisted navigation during surgery. The system includes at least one processor configured to perform operations that include to identify coordinates of a pattern of spaced-apart light reflective material areas along a continuous surface of a reference marker attached to a real device in images obtained from tracking cameras imaging the reference marker with at least partially overlapping field-of-views. The operations track pose of the reference marker in 3D space based on the identified coordinates of the pattern of

spaced-apart light reflective material areas along the continuous surface of the reference marker.

[0013] Some other embodiments of the present disclosure are directed to a reference marker configured to be tracked by a camera tracking system for computer assisted navigation during surgery. The reference marker includes an object with a surface of one of a light absorptive material and a light reflective material, and a cover layer extending across the surface of the object. The cover layer is the other one of the light absorptive material and the light reflective material. The cover layer has a pattern of openings exposing spaced-apart areas of the surface of the object.

[0014] Other camera tracking systems and reference markers according to embodiments of the inventive subject matter will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such additional camera tracking systems and reference markers be included within this description, be within the scope of the present inventive subject matter, and be protected by the accompanying claims. Moreover, it is intended that all embodiments disclosed herein can be implemented separately or combined in any way and/or combination.

DESCRIPTION OF THE DRAWINGS

[0015] Aspects of the present disclosure are illustrated by way of example and are not limited by the accompanying drawings. In the drawings:

[0016] FIG. 1 is an overhead view of a surgical system arranged during a surgical procedure in a surgical room which includes a camera tracking system for navigated surgery and which may further include a surgical robot for robotic assistance according to some embodiments;

[0017] FIG. 2 illustrates the camera tracking system and the surgical robot positioned relative to a patient according to some embodiments;

[0018] FIG. 3 further illustrates the camera tracking system and the surgical robot configured according to some embodiments;

[0019] FIG. 4 illustrates a block diagram of a surgical system that includes an XR headset, a computer platform, imaging devices, and a surgical robot which are configured to operate according to some embodiments;

[0020] FIG. 5 illustrates an instrument that can be tracked by the camera tracking system as its tip is touched to a verification divot on an arm supporting a tracked reference array which is also tracked by the camera tracking system;

[0021] FIG. 6 illustrates components of a camera tracking system which include a neural network to predict coordinate locations of a device feature in images from tracking cameras according to some embodiments;

[0022] FIG. 7 illustrates a flowchart of operations for identifying, training, testing, and deploying a neural network in a computer automated manner according to some embodiments;

[0023] FIG. 8 illustrates a top view of a camera tracking system which includes a pair of accuracy cameras used to verify an instrument or other device for computer assisted navigation during surgery according to some embodiments;

[0024] FIG. 9 illustrates another top view of the camera tracking system of FIG. 8 which includes a calibration pattern used to calibrate relative pose of the accuracy cameras according to some embodiments;

[0025] FIG. 10 illustrates a flowchart of operations that may be performed by the camera tracking system of FIG. 8 to calibrate positioning of the accuracy cameras according to some embodiments;

[0026] FIG. 11 illustrates a flowchart of operations that may be performed by the camera tracking system of FIG. 8 for computer assisted navigation during surgery according to some embodiments;

[0027] FIG. 12 illustrates a reference marker being attached to a post of an instrument or other device according to some embodiments; and

[0028] FIGS. 13, 14a, 14b, 15a, 15b, and 16 illustrate a portion of different embodiments of reference markers having a cover layer with a pattern of openings exposing spaced-apart areas of an underlying object surface for tracking by a camera tracking system.

DETAILED DESCRIPTION

[0029] It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the description herein or illustrated in the drawings. The teachings of the present disclosure may be used and practiced in other embodiments and practiced or carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “attached,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, attachments, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

[0030] The following discussion is presented to enable a person skilled in the art to make and use embodiments of the present disclosure. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the principles herein can be applied to other embodiments and applications without departing from embodiments of the present disclosure. Thus, the embodiments are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the embodiments. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of the embodiments.

[0031] Some embodiments of the present disclosure are directed to various camera tracking systems for verifying instruments and other devices at time of use and tracking pose of the instruments and other devices during computer assisted navigation during surgery. Some other embodiments are directed to reference markers which are configured to be tracked by a camera tracking system for computer assisted navigation during surgery. Before describing these embodiments in detail, various components that may be used

for performing embodiments in a navigated surgery system are described with reference to FIGS. 1-4.

[0032] FIG. 1 is an overhead view of a surgical system arranged during a surgical procedure in a surgical room which includes a camera tracking system **200** for navigated surgery and which may further include a surgical robot **100** for robotic assistance according to some embodiments. FIG. 2 illustrates the camera tracking system **200** and the surgical robot **100** positioned relative to a patient according to some embodiments. FIG. 3 further illustrates the camera tracking system **200** and the surgical robot **100** configured according to some embodiments. FIG. 4 illustrates a block diagram of a surgical system that includes an XR headset **150**, a computer platform **400**, imaging devices **420**, and the surgical robot **100** which are configured to operate according to some embodiments.

[0033] The XR headsets **150** may be configured to augment a real-world scene with computer generated XR images while worn by personnel in the operating room. The XR headsets **150** may be configured to provide an augmented reality (AR) viewing environment by displaying the computer generated XR images on a see-through display screen that allows light from the real-world scene to pass therethrough for combined viewing by the user. Alternatively, the XR headsets **150** may be configured to provide a virtual reality (VR) viewing environment by preventing or substantially preventing light from the real-world scene from being directly viewed by the user while the user is viewing the computer-generated AR images on a display screen. The XR headsets **150** can be configured to provide both AR and VR viewing environments. Thus, the term XR headset can be referred to as an AR headset or a VR headset.

[0034] Referring to FIGS. 1-4, the surgical robot **100** may include, for example, one or more robot arms **104**, a display **110**, an end-effector **112**, for example, including a guide tube **114**, and an end effector reference array which can include one or more tracking markers. A patient reference array **116** (DRB) has a plurality of tracking markers **117** and is secured directly to the patient **210** (e.g., to a bone of the patient **210**). A reference array **170** is attached or formed on an instrument, surgical tool, surgical implant device, etc.

[0035] The camera tracking system **200** includes tracking cameras **204** which may be spaced apart stereo cameras configured with partially overlapping field-of-views. The camera tracking system **200** can have any suitable configuration of arm(s) **202** to move, orient, and support the tracking cameras **204** in a desired location, and may contain at least one processor operable to track location of an individual marker and pose of an array of markers.

[0036] As used herein, the term “pose” refers to the location (e.g., along 3 orthogonal axes) and/or the rotation angle (e.g., about the 3 orthogonal axes) of markers (e.g., DRB) relative to another marker (e.g., surveillance marker) and/or to a defined coordinate system (e.g., camera coordinate system). A pose may therefore be defined based on only the multidimensional location of the markers relative to another marker and/or relative to the defined coordinate system, based on only the multidimensional rotational angles of the markers relative to the other marker and/or to the defined coordinate system, or based on a combination of the multidimensional location and the multidimensional rotational angles. The term “pose” therefore is used to refer to location, rotational angle, or combination thereof.

[0037] The tracking cameras **204** may include, e.g., infrared cameras (e.g., bifocal or stereophotogrammetric cameras), operable to identify, for example, active and passive tracking markers for single markers (e.g., surveillance marker) and reference arrays which can be formed on or attached to the patient **210** (e.g., patient reference array, DRB), end effector **112** (e.g., end effector reference array), XR headset(s) **150** worn by a surgeon **120** and/or a surgical assistant **126**, etc. in a given measurement volume of a camera coordinate system while viewable from the perspective of the tracking cameras **204**. The tracking cameras **204** may scan the given measurement volume and detect light that is emitted or reflected from the markers in order to identify and determine locations of individual markers and poses of the reference arrays in three-dimensions. For example, active reference arrays may include infrared-emitting markers that are activated by an electrical signal (e.g., infrared light emitting diodes (LEDs)), and passive reference arrays may include retro-reflective markers that reflect infrared light (e.g., they reflect incoming IR radiation into the direction of the incoming light), for example, emitted by illuminators on the tracking cameras **204** or other suitable device.

[0038] The XR headsets **150** may each include tracking cameras (e.g., spaced apart stereo cameras) that can track location of a surveillance marker and poses of reference arrays within the XR camera headset field-of-views (FOVs) **152** and **154**, respectively. Accordingly, as illustrated in FIG. 1, the location of the surveillance marker and the poses of reference arrays on various objects can be tracked while in the FOVs **152** and **154** of the XR headsets **150** and/or a FOV **600** of the tracking cameras **204**.

[0039] FIGS. 1 and 2 illustrate a potential configuration for the placement of the camera tracking system **200** and the surgical robot **100** in an operating room environment. Computer-aided navigated surgery can be provided by the camera tracking system controlling the XR headsets **150** and/or other displays **34**, **36**, and **110** to display surgical procedure navigation information. The surgical robot **100** is optional during computer-aided navigated surgery.

[0040] The camera tracking system **200** may operate using tracking information and other information provided by multiple XR headsets **150** such as inertial tracking information and optical tracking information (frames of tracking data). The XR headsets **150** operate to display visual information and may play-out audio information to the wearer. This information can be from local sources (e.g., the surgical robot **100** and/or other medical), remote sources (e.g., patient medical image server), and/or other electronic equipment. The camera tracking system **200** may track markers in 6 degrees-of-freedom (6 DOF) relative to three axes of a 3D coordinate system and rotational angles about each axis. The XR headsets **150** may also operate to track hand poses and gestures to enable gesture-based interactions with “virtual” buttons and interfaces displayed through the XR headsets **150** and can also interpret hand or finger pointing or gesturing as various defined commands. Additionally, the XR headsets **150** may have a 1-10× magnification digital color camera sensor called a digital loupe. In some embodiments, one or more of the XR headsets **150** are minimalistic XR headsets that display local or remote information but include fewer sensors and are therefore more lightweight.

[0041] An “outside-in” machine vision navigation bar supports the tracking cameras **204** and may include a color

camera. The machine vision navigation bar generally has a more stable view of the environment because it does not move as often or as quickly as the XR headsets **150** while positioned on wearers' heads. The patient reference array **116** (DRB) is generally rigidly attached to the patient with stable pitch and roll relative to gravity. This local rigid patient reference **116** can serve as a common reference for reference frames relative to other tracked arrays, such as a reference array on the end effector **112**, instrument reference array **170**, and reference arrays on the XR headsets **150**.

[0042] During a surgical procedure using surgical navigation, a surveillance marker can be affixed to the patient to provide information on whether the patient reference array **116** has shifted. For example, during a spinal fusion procedure with planned placement of pedicle screw fixation, two small incisions are made over the posterior superior iliac spine bilaterally. The DRB and the surveillance marker are then affixed to the posterior superior iliac spine bilaterally. If the surveillance marker's location changes relative to the patient reference array **116**, the camera tracking system **200** may display a meter indicating the amount of movement and/or may display a pop-up warning message to inform the user that the patient reference array may have been bumped. If the patient reference array has indeed been bumped, the registration of the patient reference array to the tracked coordinate system may be invalid and could result in erroneous navigation which is off target.

[0043] When present, the surgical robot (also "robot") may be positioned near or next to patient **210**. The robot **100** can be positioned at any suitable location near the patient **210** depending on the area of the patient **210** undergoing the surgical procedure. The camera tracking system **200** may be separated from the robot system **100** and positioned at the foot of patient **210**. This location allows the tracking camera **200** to have a direct visual line of sight to the surgical area **208**. In the configuration shown, the surgeon **120** may be positioned across from the robot **100**, but is still able to manipulate the end-effector **112** and the display **110**. A surgical assistant **126** may be positioned across from the surgeon **120** again with access to both the end-effector **112** and the display **110**. If desired, the locations of the surgeon **120** and the assistant **126** may be reversed. An anesthesiologist **122**, nurse or scrub tech can operate equipment which may be connected to display information from the camera tracking system **200** on a display **34**.

[0044] With respect to the other components of the robot **100**, the display **110** can be attached to the surgical robot **100** or in a remote location. End-effector **112** may be coupled to the robot arm **104** and controlled by at least one motor. In some embodiments, end-effector **112** can comprise a guide tube **114**, which is configured to receive and orient a surgical instrument, tool, or implant used to perform a surgical procedure on the patient **210**.

[0045] As used herein, the term "end-effector" is used interchangeably with the terms "end-effector" and "effector element." The term "instrument" is used in a non-limiting manner and can be used interchangeably with "tool" and "implant" to generally refer to any type of device that can be used during a surgical procedure in accordance with embodiments disclosed herein. The more general term device can also refer to structure of the end-effector, etc. Example instruments, tools, and implants include, without limitation, drills, screwdrivers, saws, dilators, retractors, probes, implant inserters, and implant devices such as a

screws, spacers, interbody fusion devices, plates, rods, etc. Although generally shown with a guide tube **114**, it will be appreciated that the end-effector **112** may be replaced with any suitable instrumentation suitable for use in surgery. In some embodiments, end-effector **112** can comprise any known structure for effecting the movement of the surgical instrument in a desired manner.

[0046] The surgical robot **100** is operable to control the translation and orientation of the end-effector **112**. The robot **100** may move the end-effector **112** under computer control along x-, y-, and z-axes, for example. The end-effector **112** can be configured for selective rotation about one or more of the x-, y-, and z-axis, and a Z Frame axis, such that one or more of the Euler Angles (e.g., roll, pitch, and/or yaw) associated with end-effector **112** can be selectively computer controlled. In some embodiments, selective control of the translation and orientation of end-effector **112** can permit performance of medical procedures with significantly improved accuracy compared to conventional robots that utilize, for example, a 6 DOF robot arm comprising only rotational axes. For example, the surgical robot **100** may be used to operate on patient **210**, and robot arm **104** can be positioned above the body of patient **210**, with end-effector **112** selectively angled relative to the z-axis toward the body of patient **210**.

[0047] In some example embodiments, the XR headsets **150** can be controlled to dynamically display an updated graphical indication of the pose of the surgical instrument so that the user can be aware of the pose of the surgical instrument at all times during the procedure.

[0048] In some further embodiments, surgical robot **100** can be operable to correct the path of a surgical instrument guided by the robot arm **104** if the surgical instrument strays from the selected, preplanned trajectory. The surgical robot **100** can be operable to permit stoppage, modification, and/or manual control of the movement of end-effector **112** and/or the surgical instrument. Thus, in use, a surgeon or other user can use the surgical robot **100** as part of computer assisted navigated surgery, and has the option to stop, modify, or manually control the autonomous or semi-autonomous movement of the end-effector **112** and/or the surgical instrument.

[0049] Reference arrays of markers can be formed on or connected to robot arms **102** and/or **104**, the end-effector **112** (e.g., end-effector array **114** in FIG. 2), and/or a surgical instrument (e.g., instrument array **170**) to track poses in 6 DOF along 3 orthogonal axes and rotation about the axes. The reference arrays enable each of the marked objects (e.g., the end-effector **112**, the patient **210**, and the surgical instruments) to be tracked by the tracking camera **200**, and the tracked poses can be used to provide navigated guidance during a surgical procedure and/or used to control movement of the surgical robot **100** for guiding the end-effector **112** and/or an instrument manipulated by the end-effector **112**.

[0050] Referring to FIG. 3 the surgical robot **100** may include a display **110**, upper arm **102**, lower arm **104**, end-effector **112**, vertical column **312**, casters **314**, a table **318**, and ring **324** which uses lights to indicate statuses and other information. Cabinet **106** may house electrical components of surgical robot **100** including, but not limited, to a battery, a power distribution module, a platform interface board module, and a computer. The camera tracking system

200 may include a display **36**, tracking cameras **204**, arm(s) **202**, a computer housed in cabinet **330**, and other components.

[0051] In computer-assisted navigated surgeries, perpendicular 2D scan slices, such as axial, sagittal, and/or coronal views, of patient anatomical structure are displayed to enable user visualization of the patient's anatomy alongside the relative poses of surgical instruments. An XR headset or other display can be controlled to display one or more 2D scan slices of patient anatomy along with a 3D graphical model of anatomy. The 3D graphical model may be generated from a 3D scan of the patient, e.g., by a CT scan device, and/or may be generated based on a baseline model of anatomy which isn't necessarily formed from a scan of the patient.

Example Surgical System

[0052] FIG. 4 illustrates a block diagram of a surgical system that includes an XR headset **150**, a computer platform **400**, imaging devices **420**, and a surgical robot **100** which are configured to operate according to some embodiments.

[0053] The imaging devices **420** may include a C-arm imaging device, an O-arm imaging device, and/or a patient image database. The XR headset **150** provides an improved human interface for performing navigated surgical procedures. The XR headset **150** can be configured to provide functionalities, e.g., via the computer platform **400**, that include without limitation any one or more of: identification of hand gesture based commands, display XR graphical objects on a display device **438** of the XR headset **150** and/or another display device. The display device **438** may include a video projector, flat panel display, etc. The user may view the XR graphical objects as an overlay anchored to particular real-world objects viewed through a see-through display screen. The XR headset **150** may additionally or alternatively be configured to display on the display device **438** video streams from cameras mounted to one or more XR headsets **150** and other cameras.

[0054] Electrical components of the XR headset **150** can include a plurality of cameras **430**, a microphone **432**, a gesture sensor **434**, a pose sensor (e.g., inertial measurement unit (IMU)) **436**, the display device **438**, and a wireless/wired communication interface **440**. The cameras **430** of the XR headset **150** may be visible light capturing cameras, near infrared capturing cameras, or a combination of both.

[0055] The cameras **430** may be configured to operate as the gesture sensor **434** by tracking for identification user hand gestures performed within the field-of-view of the camera(s) **430**. Alternatively, the gesture sensor **434** may be a proximity sensor and/or a touch sensor that senses hand gestures performed proximately to the gesture sensor **434** and/or senses physical contact, e.g., tapping on the sensor **434** or its enclosure. The pose sensor **436**, e.g., IMU, may include a multi-axis accelerometer, a tilt sensor, and/or another sensor that can sense rotation and/or acceleration of the XR headset **150** along one or more defined coordinate axes. Some or all of these electrical components may be contained in a head-worn component enclosure or may be contained in another enclosure configured to be worn elsewhere, such as on the hip or shoulder.

[0056] As explained above, a surgical system includes the camera tracking system **200** which may be connected to a computer platform **400** for operational processing and which

may provide other operational functionality including a navigation controller **404** and/or of an XR headset controller **410**. The surgical system may include the surgical robot **100**. The navigation controller **404** can be configured to provide visual navigation guidance to an operator for moving and positioning a surgical tool relative to patient anatomical structure based on a surgical plan, e.g., from a surgical planning function, defining where a surgical procedure is to be performed using the surgical tool on the anatomical structure and based on a pose of the anatomical structure determined by the camera tracking system **200**. The navigation controller **404** may be further configured to generate navigation information based on a target pose for a surgical tool, a pose of the anatomical structure, and a pose of the surgical tool and/or an end effector of the surgical robot **100**, where the steering information is displayed through the display device **438** of the XR headset **150** and/or another display device to indicate where the surgical tool and/or the end effector of the surgical robot **100** should be moved to perform the surgical plan.

[0057] The electrical components of the XR headset **150** can be operatively connected to the electrical components of the computer platform **400** through the wired/wireless interface **440**. The electrical components of the XR headset **150** may be operatively connected, e.g., through the computer platform **400** or directly connected, to various imaging devices **420**, e.g., the C-arm imaging device, the I/O-arm imaging device, the patient image database, and/or to other medical equipment through the wired/wireless interface **440**.

[0058] The surgical system may include a XR headset controller **410** that may at least partially reside in the XR headset **150**, the computer platform **400**, and/or in another system component connected via wired cables and/or wireless communication links. Various functionality is provided by software executed by the XR headset controller **410**. The XR headset controller **410** is configured to receive information from the camera tracking system **200** and the navigation controller **404**, and to generate an XR image based on the information for display on the display device **438**.

[0059] The XR headset controller **410** can be configured to operationally process frames of tracking data from tracking cameras from the cameras **430** (tracking cameras), signals from the microphone **1620**, and/or information from the pose sensor **436** and the gesture sensor **434**, to generate information for display as XR images on the display device **438** and/or as other for display on other display devices for user viewing. Thus, the XR headset controller **410** illustrated as a circuit block within the XR headset **150** is to be understood as being operationally connected to other illustrated components of the XR headset **150** but not necessarily residing within a common housing or being otherwise transportable by the user. For example, the XR headset controller **410** may reside within the computer platform **400** which, in turn, may reside within the cabinet **330** of the camera tracking system **200**, the cabinet **106** of the surgical robot **100**, etc.

Camera Tracking System With Neural Network Prediction of Device Feature Coordinate Locations

[0060] Some embodiments are directed to various camera tracking systems which are configured to directly track a tip of an instrument or other feature of a device using navigation cameras for verification, calibration, and/or navigation during surgery. Although various embodiments are described in

the context of tracking an instrument tip, they are not limited thereto and can be used to track any visually recognizable feature of a real device which is modeled in a computer model (e.g., computer aided drawing) of the real device. The real device may correspond to an instrument, end effector, or other apparatus.

[0061] FIG. 6 illustrates components of a camera tracking system which include a neural network configured to predict coordinate locations of a device feature in images from tracking cameras according to some embodiments. Referring to FIG. 6, the camera tracking system 200 includes tracking cameras 600 such as the cameras 204 in FIG. 1, an image processing module 602, a neural network 604 such as a convolutional neural network, and an instrument verification and tracking module 606.

[0062] In accordance with some embodiments, the image processing module 602 identifies locations of markers of a reference array in a set of the images obtained from the tracking cameras 600 imaging a real device with at least partially overlapping field-of-views. The image processing module 602 determines measured coordinate locations of a feature of a real device in the set of the images based on the identified locations of the markers and based on a relative location relationship between the markers and the feature. The system 200 processes a region of interest in the set of the images identified based on the measured coordinate locations through the neural network 604 which is configured to output a prediction of coordinate locations of the feature in the set of the images. The neural network 604 has been trained based on training images containing the feature of a computer model rendered at known coordinate locations. The instrument verification and tracking module 606 tracks pose of the feature of the real device in three-dimensional (3D) space based on the prediction of coordinate locations of the feature of the real device in the set of the images.

[0063] In a further embodiment, the prediction of coordinate locations of the feature of the real device, based on output of the neural network 604, indicates predicted pixel coordinates of the feature within the set of the images.

[0064] In a further embodiment, the instrument verification and tracking module 606 determines a predicted 3D pose of the feature of the real device in a tracked space based on triangulation of the prediction of two-dimensional (2D) coordinate locations of the feature of the real device in a pair of the set of the images from a pair of the tracking cameras 600. The instrument verification and tracking module 606 determines a measured 3D pose of the feature of the real device in the tracked space based on triangulation of the locations of the markers of the reference array in the pair of the set of the images, and calibrates a feature offset based on comparison of the predicted 3D pose of the feature and the measured 3D pose of the feature.

[0065] In order to track the real feature(s), also called datum(s), of the real device datums, the computer model (e.g., known CAD data) can be used to train the neural network 604 to return the image pixel coordinates of the feature (datum) requested. In an example scenario, the high contrast optical markers can have tracked locations in the same image frame as an instrument tip to allow operations to calibrate the markers and instrument tip into the same rigid body. In this way, the need for mechanical “verification divots” is eliminated and real-time and automated measurement of the instrumentation and apparatus can be operation-

ally performed. After finding the tip location in left and right stereo cameras, the 3D location is determined using triangulation of the 2D locations. The markers can be similarly identified in 2D imagery before being triangulated to determine their 3D locations.

[0066] In a further embodiment, the instrument verification and tracking module 606 verifies the feature of the real device based on whether the measured coordinate locations of the feature of the real device are within a threshold distance of the prediction of coordinate locations of the feature of the real device.

[0067] In a further embodiment, training images, which contain the feature of the computer model rendered at the known coordinate locations in the training images, are processed through the neural network 604 to output predictions of coordinate locations of the feature of the computer model in the training images. The known coordinate locations of the feature in the training images are compared to the predictions of coordinate locations of the feature of the computer model in the training images, and parameters of the neural network 604 are trained based on the comparison.

[0068] The neural network 604 may be trained using, for example, tens of thousands or millions of images. An alpha image can be generated with a computer generated representation of the device rendering based on the computer model, and which may be rendered with a transparent background. The alpha image can then be augmented with different synthetic backgrounds to generate the training images for use in training the neural network 604. In one embodiment, the training images are generated to contain the feature of the computer model rendered at the known coordinate locations in the training images with different rendered backgrounds, different rendered lighting conditions, and/or different rendered poses of the feature of the computer model between at least some of the training images.

[0069] The operation to train the parameters of the neural network 604 may include to adapt weights and/or firing thresholds assigned to combining nodes of at least one layer of the neural network 604, based on the training images containing the feature of the computer model rendered at the known coordinate locations in the training images.

[0070] A convolutional neural network 604 (CNN) can be used which is configured to rule in real-time or near-real-time processing images from the tracking cameras 600. CNNs are also known as Shift Invariant or Space Invariant Artificial Neural Networks (SIANN), based on the shared-weight architecture of the convolution kernels or filters that slide along image input features and provide translation-equivariant responses known as feature maps.

[0071] In a CNN, the input is a tensor with a shape: (number of inputs) $\times$ (input height) $\times$ (input width) $\times$ (input channels). After passing through a convolutional layer, the image becomes abstracted to a feature map, also called an activation map, with shape: (number of inputs) $\times$ (feature map height) $\times$ (feature map width) $\times$ (feature map channels). Convolutional layers convolve the input and pass its result to the next layer.

[0072] Rather than train one neural network to track a plurality of different types of instruments (which may require an excessively large architecture and be very time consuming), in some embodiments a different neural network is trained and used to predict feature coordinate locations of different types of devices. Effective training can

in at least some scenarios require thousands of labeled images used for supervised training operations. Training hundreds of relatively small device-type-dedicated neural networks to correspond to hundreds of different types of instruments and other apparatus may be extremely time consuming to the point of being impractical. Accordingly, some embodiments are directed to automating the training, evaluation and deployment via simulator training operations. The simulator can include a photosphere (optionally HORI) with dynamic lighting. The relevant instrument is added into the scene and virtual navigation cameras with very similar lens characteristics capture the reference element and instrument tip in thousands of different poses with widely varying backgrounds. The output images are then processed at the time of training in order to match the navigation cameras themselves. For example, the images may be made monochrome, a brightness constant is applied, noise may be applied with a blur kernel.

[0073] Once several thousand to more than one million simulated images are fully augmented, the neural network is trained using the simulated images. Evaluating performance of the neural network involves testing the neural network against thousands of varying test images that are not in the training dataset (i.e., have not been seen by the neural network before). Depending on the result of comparing predicted measurements versus known ground truth, the neural network is either deployed as being successful (for use in operating rooms in accordance with embodiments herein) or failed and sent back for continued training improvement. The neural network may be fast enough that an ensemble of several to dozens of neural network can be trained and run in the field. If the variance between neural network is excessive, the measurement is rejected. Averaging the ensemble output may also improve accuracy by approximately 10% to 30%.

[0074] FIG. 7 illustrates a flowchart of operations for identifying, training, testing, and deploying a neural network in an computer automated manner according to some embodiments.

[0075] Referring to FIG. 7, a computer model of a new instrument type is configured **700**, which may include designing a 3D model of the instrument in a computer-aided design tool. A computer generated image of the tool is rendered **702** in training images with a variety of differing backgrounds and lighting conditions. A neural network is trained (i.e., parameters are trained) and tested **704** using the training images. The testing accuracy is determined **706** and, if the testing accuracy is within an acceptable range, the trained neural network for the new tool type is integrated **708** into a camera tracking system. The neural network can then be executed **710** during a surgical procedure on live image data to verify accuracy of a feature (datum) of the real instrument corresponding to the new instrument type, and to track location of the feature (datum) of the real instrument.

[0076] In contrast, when the test accuracy is determined **706** to not be within the acceptable range, the operations evaluate **712** the results and a determination **714** is made whether a rule is satisfied indicating that test accuracy for the new instrument can be improved through further training of the neural network. If the rule is satisfied, operations adapt **716** the neural network and/or the training images data (e.g., varying the backgrounds and lighting conditions) based on the evaluation **712**, and the neural network is further trained

and tested **704**. In contrast, when the rule is not satisfied, the operations indicate **718** rejection of support for tracking the requested feature (datum).

Camera Tracking System With Accuracy Cameras for Device Feature Verification

[0077] In the event that the navigation cameras are not well calibrated or operating room conditions are too uncontrolled and dynamic, other operations can be performed using a specialized verification and calibration fixture attached to visible light optical cameras (also called “accuracy cameras”). The calibration fixture and associated operations can be used an alternative approach to the above-approach of a neural network-based verification of instruments, or may be used in combination with the above-approach.

[0078] The operations can use computer vision techniques which are applied to the accuracy cameras working in tandem with existing near-infrared (NIR) tracking cameras **200** (FIG. 2). The computer vision techniques can also make use of neural networks such as any of the embodiments described above.

[0079] The operations can provide meaningful visual feedback to the user that includes live feeds of the accuracy cameras video data with diagnostic overlay. When used with an XR headset, the display can present a floating window including either left or right tracking cameras and/or accuracy cameras with a computer-rendered overlay of predicted versus actual computer model, e.g., computer-aided design (CAD), information. Visual guidance to the user can also be shown (e.g., via arrows) directing the user to move the instrument or other device one way or another for improved measurement perspective relative to the accuracy cameras. Similarly, a 3D model can be attached to the end of the instrument or other device for spatial awareness. This information can be to scale or enlarged for enhanced visibility.

[0080] Live or paused image or video information can be displayed during or after the verification and/or calibration process on 2D monitor(s). Graphical overlay of measured versus expected information including numbers or other diagnostics can be shown.

[0081] In either case, the visualization can update in real time as the instrument and/or other device is moved around, providing intuitive feedback independent of automated measurements. In cases such as incorrect instrument or implant attachments the error may become immediately and intuitively known to surgical staff from these visualizations.

[0082] In some embodiments, multiple views of a tracked instrument are captured with the accuracy cameras that are themselves tracked by tracking cameras **200**. The instruments and the accuracy cameras include reference elements that are tracked by the tracking cameras **200**.

[0083] FIG. 8 illustrates a top view of a camera tracking system which includes a pair of accuracy cameras **800** used to verify an instrument or other device for computer assisted navigation during surgery according to some embodiments. The accuracy cameras **800** are connected to a reference array **812** of markers through a rigid fixture **810** to eliminate any movement between the accuracy cameras **800**. The accuracy cameras **800** are oriented by the fixture **810** to have at least partially overlapping field-of-views of a verification space **820** in which an instrument feature can be located during verification operations. The accuracy cameras **800** may have higher pixel resolution but have a narrower field-of-view

than the tracking cameras **200** since the instrument feature is positioned relatively close in the verification space **820** to the accuracy cameras **800**.

[0084] A screen **830** may extend from the fixture **810** to be located behind the verification space **820** relative to the accuracy cameras **800**. The screen **830** can have a steady or adjustable color and/or provide lighting, e.g., electroluminescent panel, to create a desired background to an instrument being verified while imaged by the accuracy cameras **800**. The desired background can improve repeatability, reliability and accuracy of the computer vision techniques when determining 3D location of the instrument feature in images from the accuracy cameras **800**.

[0085] The accuracy cameras **800** may be covered in a thin sterile transparent drape that is pulled taught (or made from a hard shell) so as to not impact the cameras visibility or add distortion or refraction.

[0086] A calibration pattern may be used to calibrate the relative pose between the accuracy cameras **800**. FIG. **9** illustrates another top view of the camera tracking system of FIG. **8** which includes a calibration pattern **900** that can be used to calibrate relative pose of the accuracy cameras **800** according to some embodiments. The calibration pattern **900** is displayed in the verification space **820** so it is contained in images obtained from the accuracy cameras **800**. Operations identify the actual calibration pattern in images from accuracy cameras **800**, and compute an expected calibration pattern in images from accuracy cameras **800** based on a relative pose of the accuracy cameras **800**. Operations then adjust the relative pose of the accuracy cameras **800** based on difference between the actual calibration pattern and the expected calibration pattern in the images from the accuracy cameras **800**. The adjustment of the relative pose may include adjusting a predefined relationship characterizing the distance and angular offset between the accuracy cameras **800**.

[0087] FIG. **10** illustrates a flowchart of operations that may be performed by the camera tracking system of FIG. **8** to calibrate positioning of the accuracy cameras **800** according to some embodiments.

[0088] Referring to FIG. **10**, the operations estimate **1002** the relative pose (e.g.,) between the accuracy cameras **800**, e.g., based on a defined baseline relative pose according to accuracy cameras **800** and the fixture **810**. The operations compute **1000** an expected calibration pattern that is expected to be captured in images from the accuracy cameras **800** based on the defined pattern of the calibration pattern and based on the estimate of the relative pose. Operations compute **1004** a difference between the expected calibration pattern and an actual calibration pattern identified in the images from the accuracy cameras **800**. When the difference is determined **1006** to satisfy a defined rule, such as by being less than a threshold error, the relative pose between the accuracy cameras **800** is calibrated **1010** based on the then-existing estimate. In contrast, when the difference is determined **1006** to not satisfy the defined rule, the estimate of the relative pose is adjusted **1008** and operations are repeated to compute **1000** another expected pattern, compute **1004** the difference, and to determine **1006** whether the difference satisfies the defined rule.

[0089] FIG. **11** illustrates a flowchart of operations that may be performed by the camera tracking system of FIG. **8** for computer assisted navigation during surgery according to some embodiments.

[0090] Referring to FIG. **11**, the operations estimate **1100** pose of a feature, e.g., tip, of an instrument based on coordinate locations of the feature identified in a set of images obtained from the accuracy cameras. The operations determine **1104** an expected pose of the instrument feature. The determination **1104** can include to measure pose of markers of a reference array attached to the instrument in a set of images obtained from tracking cameras imaging the reference array and the instrument with at least partially overlapping field-of-views. The determination **1104** further includes to determine expected pose of the feature of the instrument based on a computer model **1102** of the instrument and feature, and the measured pose of the markers of the reference array. In some embodiments, the determination **1104** of the expected pose of the instrument feature may be performed using a neural network to estimate location of the instrument feature in the images from the accuracy cameras **800** and which can be translated to a 3D pose using similar operations to those described for one or more of the embodiments above.

[0091] The operations adjust a calibrated pose of the feature of the instrument based on difference between the estimated pose and the expected pose of the feature of the instrument. The adjustment operation can include to compute **1106** a difference between the the estimated pose and the expected pose of the instrument feature. When the difference does not satisfy a defined rule (e.g., feature verification rule), such as by being less than a threshold error, the calibrated pose of the instrument feature can be adjusted **1110** and/or an indication that the instrument feature is unverified can be generated (e.g., displayed to a user), and operations are repeated to determine **1104** an expected pose of the instrument feature, compute **1106** the difference, and to determine **1108** whether the difference satisfies the defined rule. When the difference is determined **1108** to satisfy the defined rule, the calibrated pose of the instrument feature can be output **1112** for use in tracking pose of the instrument feature using the tracking cameras and/or an indication that the instrument feature is verified can be generated (e.g., displayed to a user). When determined **1108** to satisfy the defined rule, the operations can allow tracking of pose of the instrument feature based on pose of the markers of the reference array attached to the instrument in images obtained from the tracking cameras and based on the calibrated pose of the instrument feature.

[0092] In one specific implementation of FIG. **11**, a system for calibrating a surgical device includes a tracking cameras system **200** having a plurality of tracking cameras **204** for tracking navigated surgical devices, a calibration fixture **810** having reference array markers **218** attached thereto and accuracy cameras **800** arranged orthogonally to each other to have at least partially overlapping field-of-views and configured to obtain images of a surgical device (see FIG. **5**, for example) having reference array markers such as shown as element **170** in FIG. **1**, and a processor for performing the surgical device calibration.

[0093] A surgical device needing calibration is placed in the calibration fixture **810**. The processor then images the surgical device and performs the following operations.

[0094] It estimates pose (e.g., 3D position and orientation) of a physical feature (such as a tip) of the surgical device based on coordinate locations of the feature identified in the set of images obtained from the accuracy cameras **800** and determine pose of the calibration fixture **810** relative to the

tracking cameras **204** based on the calibration fixture reference array markers **812** seen by the tracking cameras. Based on the above operations, the processor determines the pose of the physical feature of the surgical device relative to the tracking cameras.

[0095] The process further determine expected pose of the surgical device feature based on a computer model of the surgical device and the measured pose of the reference array markers of the surgical device contained in a set of images obtained from the tracking cameras **204**. The processor then adjusts pose of the surgical device feature in the computer model based on a difference between the estimated pose and the expected pose of the surgical device feature.

[0096] The focal length of the accuracy cameras which need to track just the surgical device being calibrated is typically very small (e.g., in the 10 cm to 60 cm range) relative to the large focal length (e.g., greater than 100 cm) of the tracking cameras which are used to track all navigated tools and instruments in an operating room.

Optical Tracking of Reference Marker Using Negative Spaces (Exposed Contrasting Recessed Surfaces)

[0097] Surgical navigation using reference markers that are tracked by tracking cameras has become a desired technique in the operating room. The reference elements are identified using unique geometry patterns created by optical markers. Active optical markers typically emit infrared light in response to a trigger emitted by an optical camera. Passive optical markers glow bright in the spectrum band of interest, typically infra-red, while the rest of the scene is relatively dark.

[0098] Two technologies for passive markers include spheres and circular disks. A spherical marker can be mounted on a post which allows wide-angle tracking due to the spherical shape and which can be relatively simple to assemble. FIG. 12 illustrates a spherical marker **1200** (reference marker) being attached to a post **1210** of an instrument or other device according to some embodiments. The marker **1200** can include a socket extended into the marker **1200** and adapted to releasably connect the a post **1210** of the instrument, other device, or reference frame. However, a spherical marker can be prone to manufacturing defects that create an imperfect sphere, have high disposable costs since they are not reusable between surgeries, and can become compromised in performance if foreign material occludes a reflective surface of the spherical marker from being imaged. In contrast, a circular disk marker can be mounted in a socket. A circular disk marker can be easier to manufacture than a spherical marker, but still have disposable costs and have a more limited angle of view compared to a spherical marker, and can become compromised in performance if foreign material occludes a reflective surface of the circular disk marker from being imaged.

[0099] Reusable optical markers can have a reflective surface which is protected by a protective lens covering that allows it to be wiped down intraoperatively in order to clean off debris.

[0100] Various further embodiments of the present disclosure are directed to a reference marker which is configured to be tracked by a camera tracking system for computer assisted navigation during surgery. The reference marker includes an object (e.g., spherical marker or disk marker) with a surface of one of a light absorptive material and a

light reflective material, and a cover layer that extends across the surface of the object. The cover layer is the other one of the light absorptive material and the light reflective material. The cover layer has a pattern of openings exposing spaced-apart areas of the surface of the object. The openings can be formed as cut-outs through the cover layer and can have any shape. The exposed spaced-apart areas of the surface of the underlying object form negative spaces that can be tracked using the tracking cameras.

[0101] FIGS. 13, 14a, 14b, 15a, 15b, and 16 illustrate a portion of different embodiments of reference markers having a cover layer with a pattern of openings exposing spaced-apart areas of an underlying object surface for tracking by a camera tracking system.

[0102] Referring initially to FIG. 13, a top view and a corresponding side-cross-sectional view show the cover layer **1300** with a pattern of openings **1310** exposing spaced-apart areas of an underlying object **1302**. The object **1302** may be a sphere and the cover layer **1300** can extend across the surface of the sphere. Alternatively, the object **1302** may be a circular disk and the cover layer **1300** can extend across the surface of the circular disk.

[0103] In the embodiment illustrated in FIG. 13, the surface of the object **1302** is a light reflective material, such as anodized aluminum. To provide optical contrast, the cover layer is then a light absorptive material, such as a black-anodized material. In another embodiment, the surface of the object **1302** is a light absorptive material and the cover layer is a light reflective material. An example thickness of the cover layer **1300** can be 200 microns, although the cover layer **1300** may be any thickness such as to provide sufficient depth to obscure one of the spaced-apart areas from being imaged by one or more of the tracking cameras when viewed sufficiently off-camera-axis.

[0104] The spaced-apart areas of the surface of the object **1302** exposed through the openings **1302** in the cover layer **1300** are circular shape in some embodiments. In other embodiments the spaced-apart areas can have any other shape, including oval, square, rectangle, etc.

[0105] The opaque cover layer **1300** may have a sufficient thickness to obscure some of the patterned spaced-apart areas of the surface of the object **1302** (negative spaces) when imaged by the tracking cameras at an off-axis angle relative to the line-of-sight of the cameras (e.g., cameras **204** in FIGS. 1 and 2). Some further embodiments are directed to avoiding obscuration of the patterned spaced-apart areas of the surface of the object **1302** when imaged off-axis angle by forming the openings in the cover layer to have sloping sidewall surfaces.

[0106] FIG. 14a is a side cross-sectional view of a portion of another reference marker which includes a cover layer **1400** with a pattern of openings **1410** exposing spaced-apart areas of an underlying object **1402**, which is configured in accordance with one embodiment. The object **1402** may be a sphere or circular disk, and the cover layer **1400** can extend across the surface of the sphere or circular disk. In contrast to the openings **1310** in FIG. 13, the openings **1410** in the cover layer **1400** are formed to have sloping sidewall surfaces. The sloping sidewall openings **1410** can enable wider-field-of-view of the pattern spaced-apart areas of the surface of the object **1302** by the tracking cameras without obscuring by the cover layer **1400**.

[0107] FIG. 14b is a side cross-sectional view of a portion of another reference marker which includes a cover layer

1400 with a pattern of openings **1420**, **1422**, and **1424** exposing spaced-apart areas of an underlying object **1402**, which is configured in accordance with one embodiment. The openings **1420**, **1422**, and **1424** in the cover layer **1400** are formed to have sloping sidewall surfaces and different viewing angles relative to a line-of-sight of the tracking cameras. The different viewing angles of the openings **1420**, **1422**, and **1424** can enable much narrower-field-of-view of the pattern spaced-apart areas of the surface of the object **1302** exposed through the openings **1420**, **1422**, and **1424** by the tracking cameras.

[0108] The reference marker does not need to have a flat surface. Instead, the reference marker may have any shape, and may be custom-shaped (e.g., as a sleeve) to conform to an angled or spherical surface of a device, such as an end-effector or tool.

[0109] FIG. 15a is a side cross-sectional view of a portion of another reference marker which includes a cover layer **1500** with a pattern of openings **1510**, **1512**, **1514** exposing spaced-apart areas of an underlying object **1502** having sloped surfaces, which is configured in accordance with one embodiment. FIG. 15b is a side cross-sectional view of a portion of another reference marker which includes a cover layer **1500** with a pattern of openings **1520**, **1522**, **1524** exposing spaced-apart areas of an underlying object **1502** having sloped surfaces, which is configured in accordance with one embodiment. The object **1502** may be a sphere, circular disk with raised central area, or other shape, and the cover layer **1400** can extend across the surface. The shape of the sloping sidewall openings **1410** affects the field-of-view of the pattern spaced-apart areas of the surface of the object **1302** exposed through the openings **1510**, **1512**, **1514**, **1520**, **1522**, and **1524** by the tracking cameras without obscuring by the cover layer **1400**. For example, the openings **1510**, **1512**, **1514** have a wider field-of-view than the openings **1520**, **1522**, **1524**.

[0110] In order for the pattern of openings to be registered reliably with the underlying surface of the object, it can be important for the cover layer to attach to the same place on the surface. Moreover, the cover layer should not move relative to the underlying object of the reference marker during tracking. The cover layer may be attached to the underlying surface of the object using pins that nest in holes, raised edges that nest in a slot, or kinematic mounts using magnets.

[0111] Some related other embodiments are directed to a camera tracking system for computer assisted navigation during surgery for use with reference markers having covers with openings that expose patterns of spaced-apart areas of an underlying object, such as various of the reference markers disclosed herein. The camera tracking system can be configured to operate to identify coordinates of a pattern of spaced-apart light reflective material areas along a continuous surface of a reference marker attached to a real device in images obtained from tracking cameras imaging the reference marker with at least partially overlapping field-of-views. The camera tracking system is further configured to track pose of the reference marker in 3D space based on the identified coordinates of the pattern of spaced-apart light reflective material areas along the continuous surface of the reference marker.

[0112] In some embodiments, tracking cameras can be used to calibrate out the small reference element errors introduced due to tolerances in the cover and/or mating

mechanisms. Such calibration may operate to decrease the cost of the covers while also improving accuracy for tracking. In such embodiments, the cover may be made of a flexible and disposable black plastic that conforms to the relevant surfaces of the underlying object and, e.g., snaps on tightly using compliant mechanisms in the cover design.

[0113] Referring initially to FIG. 16, a top view and a corresponding side-cross-sectional view show a cover layer **1600** with a pattern of openings **1610** exposing spaced-apart areas of an underlying object **1602**. The object **1602** may be a sphere and the cover layer **1600** can extend across the surface of the sphere. Alternatively, the object **1602** may be a circular disk and the cover layer **1600** can extend across the surface of the circular disk. The cover layer **1610** may be formed as a thin disposable film or as a reusable hard material, e.g., metal. The relative thinness of the cover layer **1600** avoids the cut-out edges of the openings obscuring view of the exposed areas of a underlying object **1602**. When the cover layer **1600** is removable, the cover layer **1600** may be attached to defined areas of a tracking plate **1604** attached or connected to the object **1602**, and/or the pattern of openings **1610** can be calibrated using image processing techniques. For example, the rectangular shaped cover layer **1600** can be registered in space relative to the tracking plate **1604**, and the pattern of openings **1610** can be registered with respect to the track plate **1604** to allow more accurate tracking of the spaced-apart areas of an underlying object **1602** and, thereby, the reference marker.

[0114] In some embodiments, when the surgical robot **100** can be configured to respond to detection of a metallic instrument inserted into the guide-tube **114** by turning off active markers to prevent unwanted motion when an instrument is in contact with the patient. This may make it necessary for the end-effector to be “active”, i.e., needing electrical power, driving up design complexity and costs. The end-effector **112** can include cut-outs forming pattern of spaced-apart light reflective material near the guide-tube **114**, which can be tracked by tracking cameras. An instrument array can include a cover that will cover the cut-outs when the instrument is inserted in the guide-tube. When the tracking camera cannot detect the end-effector **112** cut-outs near the guide-tube **114**, the system can determine that the instrument is in the guide-tube **114** and initiate responsive safety mitigations, such as stopping all motion or allowing only slow motion in a “safe” direction, e.g., away from the patient. In another embodiment, when an instrument inserted in the guide-tube **114** it displaces a cover which exposes cut-outs forming a pattern of spaced-apart light reflective material which was not visible to the camera before insertion of the instrument.

Further Definitions and Embodiments

[0115] In the above-description of various embodiments of present inventive concepts, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of present inventive concepts. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which present inventive concepts belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the

relevant art and will not be interpreted in an idealized or overly formal sense expressly so defined herein.

[0116] When an element is referred to as being “connected”, “coupled”, “responsive”, or variants thereof to another element, it can be directly connected, coupled, or responsive to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected”, “directly coupled”, “directly responsive”, or variants thereof to another element, there are no intervening elements present. Like numbers refer to like elements throughout. Furthermore, “coupled”, “connected”, “responsive”, or variants thereof as used herein may include wirelessly coupled, connected, or responsive. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Well-known functions or constructions may not be described in detail for brevity and/or clarity. The term “and/or” includes any and all combinations of one or more of the associated listed items.

[0117] It will be understood that although the terms first, second, third, etc. may be used herein to describe various elements/operations, these elements/operations should not be limited by these terms. These terms are only used to distinguish one element/operation from another element/operation. Thus, a first element/operation in some embodiments could be termed a second element/operation in other embodiments without departing from the teachings of present inventive concepts. The same reference numerals or the same reference designators denote the same or similar elements throughout the specification.

[0118] As used herein, the terms “comprise”, “comprising”, “include”, “including”, “includes”, “have”, “has”, “having”, or variants thereof are open-ended, and include one or more stated features, integers, elements, steps, components or functions but does not preclude the presence or addition of one or more other features, integers, elements, steps, components, functions or groups thereof. Furthermore, as used herein, the common abbreviation “e.g.”, which derives from the Latin phrase “exempli gratia,” may be used to introduce or specify a general example or examples of a previously mentioned item, and is not intended to be limiting of such item. The common abbreviation “i.e.”, which derives from the Latin phrase “id est,” may be used to specify a particular item from a more general recitation.

[0119] Example embodiments are described herein with reference to block diagrams and/or flowchart illustrations of computer-implemented methods, apparatus (systems and/or devices) and/or computer program products. It is understood that a block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by computer program instructions that are performed by one or more computer circuits. These computer program instructions may be provided to a processor circuit of a general purpose computer circuit, special purpose computer circuit, and/or other programmable data processing circuit to produce a machine, such that the instructions, which execute via the processor of the computer and/or other programmable data processing apparatus, transform and control transistors, values stored in memory locations, and other hardware components within such circuitry to implement the functions/acts specified in the block diagrams and/or flowchart block or blocks, and thereby create means (functionality)

and/or structure for implementing the functions/acts specified in the block diagrams and/or flowchart block(s).

[0120] These computer program instructions may also be stored in a tangible computer-readable medium that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instructions which implement the functions/acts specified in the block diagrams and/or flowchart block or blocks. Accordingly, embodiments of present inventive concepts may be embodied in hardware and/or in software (including firmware, resident software, micro-code, etc.) that runs on a processor such as a digital signal processor, which may collectively be referred to as “circuitry,” “a module” or variants thereof.

[0121] It should also be noted that in some alternate implementations, the functions/acts noted in the blocks may occur out of the order noted in the flowcharts. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Moreover, the functionality of a given block of the flowcharts and/or block diagrams may be separated into multiple blocks and/or the functionality of two or more blocks of the flowcharts and/or block diagrams may be at least partially integrated. Finally, other blocks may be added/inserted between the blocks that are illustrated, and/or blocks/operations may be omitted without departing from the scope of inventive concepts. Moreover, although some of the diagrams include arrows on communication paths to show a primary direction of communication, it is to be understood that communication may occur in the opposite direction to the depicted arrows.

[0122] Many variations and modifications can be made to the embodiments without substantially departing from the principles of the present inventive concepts. All such variations and modifications are intended to be included herein within the scope of present inventive concepts. Accordingly, the above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended examples of embodiments are intended to cover all such modifications, enhancements, and other embodiments, which fall within the spirit and scope of present inventive concepts. Thus, to the maximum extent allowed by law, the scope of present inventive concepts are to be determined by the broadest permissible interpretation of the present disclosure including the following examples of embodiments and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

What is claimed is:

1. A camera tracking system for computer assisted navigation during surgery, comprising:
 - a tracking camera system having a plurality of cameras for tracking navigated surgical devices with reference array markers, the tracking cameras having overlapping field of views;
 - a calibration fixture having reference array markers calibration markers attached thereto and accuracy cameras arranged to have overlapping field-of-views and configured to obtain images of the surgical device;
 - at least one processor operative to:
 - determine the pose of a physical feature of the surgical device based on:

estimating pose of the surgical device feature based on the feature identified in a set of images obtained from the accuracy cameras;
determining pose of the calibration box relative to the tracking cameras based on the calibration fixture reference array markers seen by the tracking cameras;
determine expected pose of the surgical device feature based on a computer model of the surgical device and the measured pose of the reference array markers of the surgical device contained in a set of images obtained from the tracking cameras; and
adjust pose of the surgical device feature in the computer model based on a difference between the estimated pose and the expected pose of the surgical device feature.
determine expected pose of the surgical device feature based on a computer model of the surgical device and the measured pose of the reference array markers of the surgical device contained in a set of images obtained from the tracking cameras; and
adjust pose of the surgical device feature in the computer model based on a difference between the estimated pose and the expected pose of the surgical device feature.

2. The camera tracking system of claim 1, wherein the processor is further operative to:
obtain images from the accuracy cameras imaging a calibration pattern;
identify an actual calibration pattern in the images from the accuracy cameras;
compute an expected calibration pattern in the images from the accuracy cameras based on a relative pose of the accuracy cameras; and
adjust the relative pose of the accuracy cameras based on difference between the actual calibration pattern and the expected calibration pattern in the images from the accuracy cameras.

3. The camera tracking system of claim 1, wherein the at least one processor is further operative to:
repeat the operation to obtain images from the accuracy cameras, to identify an actual calibration pattern in the images, to compute an expected calibration pattern in the images, and to adjust the relative pose of the accuracy cameras, until a difference between the actual calibration pattern and the expected calibration pattern in the images from the accuracy cameras satisfies a defined rule.

4. The camera tracking system of claim 3, wherein the at least one processor is further operative to:
output an indication that the feature of the device is unverified responsive to the difference between the estimated pose and the determined expected pose of the feature of the device not satisfying a feature verification rule.

5. The camera tracking system of claim 3, wherein the at least one processor is further operative to:
track pose of the feature of the device based on pose of the markers of the reference array attached to the device in images obtained from the tracking cameras and based on the calibrated pose of the feature of the device.

6. A camera tracking system for computer assisted navigation during surgery, comprising at least one processor operative to:

identify coordinates of a pattern of spaced-apart light reflective material areas along a continuous surface of a reference marker attached to a real device in images obtained from tracking cameras imaging the reference marker with at least partially overlapping field-of-views; and
track pose of the reference marker in three-dimensional (3D) space based on the identified coordinates of the pattern of spaced-apart light reflective material areas along the continuous surface of the reference marker
determine a predicted 3D pose of the feature of the real device in a tracked space based on triangulation of the prediction of two-dimensional (2D) coordinate locations of the feature of the real device in a pair of the set of the images from a pair of the tracking cameras;
determine a measured 3D pose of the feature of the real device in the tracked spaced based on triangulation of the locations of the markers of the reference array in the pair of the set of the images; and
calibrate a feature offset based on comparison of the predicted 3D pose of the feature and the measured 3D pose of the feature.

7. The camera tracking system of claim 6, wherein:
the prediction of coordinate locations of the feature of the real device indicates predicted pixel coordinates of the feature within the set of the images.

8. The camera tracking system of claim 6, wherein the at least one processor is further operative to:
verify the feature of the real device based on whether the measured coordinate locations of the feature of the real device are within a threshold distance of the prediction of coordinate locations of the feature of the real device.

9. The camera tracking system of claim 6, wherein the at least one processor is further operative to:
process training images, which contain the feature of the computer model rendered at the known coordinate locations in the training images, through the neural network to output predictions of coordinate locations of the feature of the computer model in the training images;
compare the known coordinate locations of the feature in the training images to the predictions of coordinate locations of the feature of the computer model in the training images; and
train parameters of the neural network based on the comparison.

10. The camera tracking system of claim 9, wherein the at least one processor is further operative to:
generate the training images containing the feature of the computer model rendered at the known coordinate locations in the training images with different rendered backgrounds, different rendered lighting conditions, and/or different rendered poses of the feature of the computer model between at least some of the training images.

11. The camera tracking system of claim 9, wherein training parameters of the neural network comprises to adapt weights and/or firing thresholds assigned to combining nodes of at least one layer of the neural network, based on the training images containing the feature of the computer model rendered at the known coordinate locations in the training images.